

FaceSense: Real-Time Face Recognition and Facial Emotion Analysis

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Abstract—Face recognition and facial emotion analysis are among the most important applications of Deep Learning and Computer Vision. This project presents a real-time intelligent system capable of recognizing individuals and detecting facial emotions using deep learning techniques and transfer learning.

The proposed system combines face detection, face recognition, and emotion classification through convolutional neural networks and pre-trained models. The application is deployed using Python, OpenCV, DeepFace, and Streamlit for real-time webcam interaction.

Experimental results demonstrate the effectiveness of transfer learning in improving recognition accuracy and reducing training time.

Index Terms—Deep Learning, Face Recognition, Emotion Recognition, Transfer Learning, CNN, Computer Vision, DeepFace, OpenCV

I. INTRODUCTION

Artificial Intelligence and Deep Learning have significantly transformed the field of computer vision. Among the most popular applications are face recognition and facial emotion analysis, which are widely used in security systems, healthcare, human-computer interaction, and smart applications.

Face recognition aims to identify or verify a person's identity based on facial features, while facial emotion recognition focuses on detecting emotional states such as happiness, sadness, anger, and surprise.

The objective of this project is to design and implement a real-time intelligent system capable of:

- Detecting faces using a webcam
- Recognizing known individuals
- Predicting facial emotions
- Displaying results in real time

To improve performance and reduce training complexity, transfer learning techniques are applied using pre-trained convolutional neural networks.

II. LITERATURE REVIEW

Several deep learning approaches have been proposed for facial emotion recognition.

Traditional machine learning methods relied on handcrafted features such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and Support Vector Machines (SVM). However, these approaches were limited in their ability to extract complex facial representations.

More recently, Convolutional Neural Networks (CNNs) have significantly improved facial emotion recognition performance by automatically learning hierarchical facial features.

Pre-trained architectures such as VGG16, MobileNetV2, and ResNet50 have also been successfully applied using transfer learning techniques. These models benefit from ImageNet pretraining and improve convergence speed and generalization.

Despite these advances, facial emotion recognition remains difficult because of:

- low-resolution datasets,
- class imbalance,
- noisy annotations,
- visual similarity between emotions.

In this work, we compare a CNN trained from scratch with transfer learning approaches based on MobileNetV2 and ResNet50 for real-time facial emotion recognition.

III. REAL-TIME FACE RECOGNITION SYSTEM

A. Overview

The first part of this project focuses on real-time face recognition using deep learning and computer vision techniques.

The objective is to identify known individuals through a webcam stream by comparing facial features extracted from live images with previously stored facial representations.

The proposed system was implemented using:

- Python
- OpenCV
- DeepFace
- Streamlit

The application performs real-time face detection, feature extraction, similarity comparison, and identity prediction.

B. System Workflow

The complete face recognition pipeline consists of the following steps:

- 1) Capture webcam frames
- 2) Detect facial regions
- 3) Extract facial embeddings
- 4) Compare embeddings with known faces
- 5) Predict identity
- 6) Display results in real time

C. Known Faces Dataset

The system uses a custom dataset containing images of known individuals.

Each person is stored inside a dedicated folder:

```
dataset/
  person1/
    img1.jpg
  person2/
    img1.jpg
```

The dataset is automatically loaded during application startup.

D. Face Detection

Face detection is performed using OpenCV and DeepFace.

The webcam continuously captures frames, and facial regions are extracted before recognition.

The detected face area is represented by:

$$(x, y, w, h) \quad (1)$$

where:

- x, y : top-left corner
- w : width
- h : height

E. Face Embedding Extraction

Instead of directly comparing raw images, DeepFace converts each face into a numerical embedding vector.

These embeddings represent high-level facial features learned by deep neural networks.

The project uses the FaceNet model integrated inside DeepFace.

Each face is represented by:

$$E = [e_1, e_2, \dots, e_n] \quad (2)$$

where E is the embedding vector.

F. Similarity Comparison

To recognize a person, the system compares the embedding generated from the webcam image with embeddings stored in the dataset.

Cosine distance was used as similarity metric:

$$\text{CosineDistance}(A, B) = 1 - \frac{A \cdot B}{\|A\| \times \|B\|} \quad (3)$$

A smaller distance indicates higher facial similarity.

If the distance is below a predefined threshold, the face is considered recognized.

G. Recognition Decision

The recognition process follows this logic:

- If similarity score $<$ threshold: recognized face
- Otherwise: unknown face

The predicted identity is displayed above the detected face using OpenCV.

H. Real-Time Visualization

The final system displays:

- face bounding box
- recognized name
- unknown label when no match exists

The interface was deployed using Streamlit for interactive webcam visualization.

IV. DATASET AND DATA ANALYSIS

A. FER2013 Dataset

For the facial emotion recognition task, we used the FER2013 dataset, a widely used benchmark dataset for facial expression recognition. It contains grayscale facial images of size 48×48 pixels categorized into seven emotion classes: angry, disgust, fear, happy, neutral, sad, and surprise.

The objective of this dataset is to train a model capable of classifying facial expressions based on visual facial features such as the mouth, eyes, eyebrows, and facial contours.

TABLE I: FER2013 Emotion Classes

Class	Emotion
0	Angry
1	Disgust
2	Fear
3	Happy
4	Neutral
5	Sad
6	Surprise

B. Exploratory Data Analysis

Before training the model, an exploratory data analysis was performed in order to better understand the structure and quality of the dataset. This step is important because the performance of deep learning models strongly depends on the quality and balance of the training data.

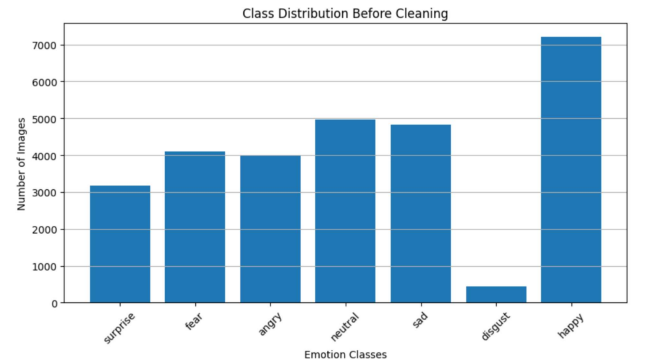


Fig. 1: Class distribution before and after preprocessing

C. Image Visualization

Several images from the FER2013 dataset were visualized in order to better understand the facial expressions, grayscale characteristics, and overall image quality.

The visualization step helped identify the diversity of facial emotions and observe the challenges associated with low-resolution grayscale images.

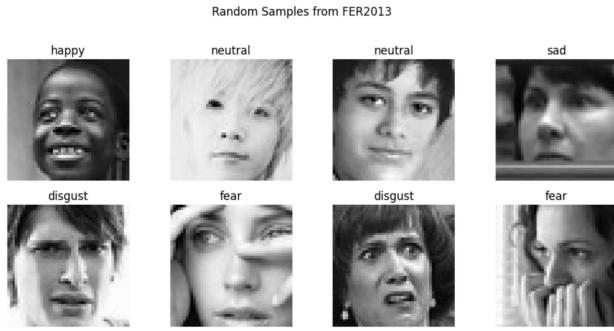


Fig. 2: Sample images from the FER2013 dataset

V. DATA PREPROCESSING

A preprocessing pipeline was applied before model training to improve data consistency and prepare the dataset for deep learning.

The preprocessing steps include:

- Image resizing
- Pixel normalization
- Data augmentation

A. Image Resizing

All images were resized to 48×48 pixels in order to match the FER2013 format and ensure compatibility with the convolutional neural network architecture.

B. Normalization

Pixel values originally range from 0 to 255. To improve model convergence during training, the pixel values were normalized into the range $[0, 1]$ using:

$$x_{norm} = \frac{x}{255} \quad (4)$$

Normalization reduces numerical instability and accelerates gradient-based optimization.

C. Data Augmentation

To improve model generalization and reduce overfitting, data augmentation techniques were applied during training.

The transformations used include:

- Horizontal flipping
- Random rotation
- Random zoom

Data augmentation artificially increases the diversity of the training data and helps the model become more robust to variations in facial appearance.

VI. CNN FROM SCRATCH

A. Model Architecture

After preprocessing, we designed a convolutional neural network from scratch for facial emotion classification. The model receives grayscale images of size $48 \times 48 \times 1$ and predicts one of the seven emotion classes.

The architecture is composed of three convolutional blocks followed by fully connected layers.

TABLE II: CNN From Scratch Architecture

Layer	Description
Input	$48 \times 48 \times 1$ grayscale image
Conv2D	32 filters, 3×3 , ReLU
BatchNormalization	Stabilizes training
MaxPooling2D	Reduces spatial dimensions
Dropout	Reduces overfitting
Conv2D	64 filters, 3×3 , ReLU
BatchNormalization	Stabilizes training
MaxPooling2D	Reduces spatial dimensions
Dropout	Reduces overfitting
Conv2D	128 filters, 3×3 , ReLU
BatchNormalization	Stabilizes training
MaxPooling2D	Reduces spatial dimensions
Dropout	Reduces overfitting
Flatten	Converts feature maps into vector
Dense	256 neurons, ReLU
Dropout	Regularization
Dense	7 neurons, Softmax

VII. TRAINING STRATEGY

The model was trained using the Adam optimizer and sparse categorical crossentropy loss function.

Since the labels are represented as integer class indices, sparse categorical crossentropy is appropriate for this multi-class classification problem.

TABLE III: Training Parameters

Parameter	Value
Optimizer	Adam
Loss Function	Sparse Categorical Crossentropy
Epochs	100
Early Stopping	Enabled
ReduceLROnPlateau	Enabled

Early stopping was used to stop training when validation loss no longer improved. The best weights were restored in order to avoid overfitting.

ReduceLROnPlateau was also applied to reduce the learning rate when the validation performance reached a plateau.

VIII. EVALUATION METRICS

To evaluate the model, several metrics were used including accuracy, precision, recall.

A. Accuracy

Accuracy measures the proportion of correctly classified samples among all samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

B. Precision

Precision measures how many predicted positive samples are actually correct:

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

C. Recall

Recall measures how many actual positive samples were correctly detected:

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

IX. RESULTS AND DISCUSSION

A. Global Results

The improved CNN from scratch achieved the following results on the FER2013 test set.

TABLE IV: Global Evaluation Results

Metric	Value
Accuracy	60.39%
Precision	60.41%
Recall	60.39%
Loss	1.058

The close values between accuracy, precision, and recall indicate that the model does not rely only on one dominant class. This suggests that class weighting helped reduce the impact of class imbalance during training.

B. Per-Class Results

TABLE V: Per-Class Classification Results

Class	Precision	Recall	F1-Score
Angry	0.50	0.53	0.52
Disgust	0.73	0.07	0.13
Fear	0.47	0.24	0.32
Happy	0.83	0.85	0.84
Neutral	0.48	0.71	0.57
Sad	0.48	0.45	0.46
Surprise	0.75	0.76	0.75

The best performance was achieved for the classes *happy* and *surprise*. These emotions are visually distinctive because happy expressions generally include a visible smile, while surprise is characterized by wide-open eyes and mouth.

However, the classes *fear* and *disgust* were more difficult to classify. The very low recall for *disgust* indicates that the model rarely predicts this class, mainly because of the limited number of training samples.

Similarly, *fear* can be confused with *sad*, *neutral*, or *surprise* because these emotions share similar facial patterns.

C. Training and Validation Analysis

The final training and validation accuracies were:

TABLE VI: Training and Validation Accuracy

Metric	Value
Training Accuracy	58.32%
Validation Accuracy	62.54%

The validation accuracy is slightly higher than the training accuracy. This behavior can be explained by the use of dropout and data augmentation during training, which make the learning process more difficult during training while improving generalization.

During validation, dropout is disabled, allowing the model to use all neurons.

The absence of a large gap between training and validation accuracy suggests that the model does not suffer from severe overfitting.

X. CONCLUSION OF THE FIRST PART

In this first part, we developed a complete pipeline for facial emotion recognition using a CNN trained from scratch.

The work included:

- exploratory data analysis,
- dataset visualization,
- preprocessing,
- data augmentation,
- CNN training,
- model evaluation.

The experimental results demonstrate that the proposed CNN achieved an accuracy of 60.39% on the FER2013 test set.

Although this result is acceptable for a CNN trained from scratch, the analysis also revealed limitations related to class imbalance and visual similarity between emotions.

Therefore, the next part of this work focuses on transfer learning using pre-trained models such as MobileNetV2 in order to improve classification performance and model generalization.

XI. TRANSFER LEARNING

After developing a CNN from scratch, we explored transfer learning approaches using pre-trained convolutional neural networks.

Transfer learning allows reusing visual knowledge learned from large-scale datasets such as ImageNet and adapting it to a new task with fewer training resources.

In this work, MobileNetV2 was selected because of:

- lightweight architecture,
- low computational cost,
- fast inference speed,
- strong performance on image classification tasks.

Unlike the CNN developed from scratch, MobileNetV2 was previously trained on millions of RGB images from the ImageNet dataset. This allows the model to learn general

visual features such as edges, textures, shapes, and object patterns before being adapted to facial emotion recognition.

A. Transfer Learning Strategy

The transfer learning pipeline consisted of the following steps:

- 1) Loading the pre-trained MobileNetV2 model
- 2) Removing the original classification head
- 3) Freezing convolutional base layers
- 4) Adding custom classification layers
- 5) Training only the newly added layers

The convolutional base was initialized with ImageNet weights using:

$$weights = ImageNet \quad (8)$$

The parameter `include_top=False` was used to remove the original ImageNet classifier.

B. Input Preprocessing

Unlike FER2013 grayscale images of size 48×48 , MobileNetV2 requires RGB images.

Therefore:

- images were converted to RGB,
- resized to 160×160 ,
- normalized using MobileNetV2 preprocessing.

The image size was reduced from 224×224 to 160×160 in order to:

- reduce GPU memory usage,
- accelerate training,
- better adapt to FER2013 low-resolution images.

C. Data Augmentation

Additional augmentation techniques were introduced in order to improve model generalization and reduce overfitting.

The augmentation pipeline includes:

- RandomFlip
- RandomRotation
- RandomZoom
- RandomTranslation
- RandomContrast

These transformations artificially increase data diversity and help the model become more robust to facial variations.

D. MobileNetV2 Architecture

The proposed architecture is illustrated in Table VII.

TABLE VII: MobileNetV2 Transfer Learning Architecture

Layer	Description
Input	$160 \times 160 \times 3$ RGB image
Data Augmentation	Random transformations
Preprocessing	MobileNetV2 preprocessing
MobileNetV2	Pre-trained convolutional base
GlobalAveragePooling2D	Feature reduction
BatchNormalization	Training stabilization
Dropout	Regularization
Dense	128 neurons, ReLU
Dropout	Regularization
Dense	7 neurons, Softmax

E. Training Configuration

The model was trained using:

- Adam optimizer,
- sparse categorical crossentropy,
- batch size = 32,
- early stopping,
- ReduceLROnPlateau callbacks.

F. Transfer Learning Results

The initial transfer learning model achieved the following results on the FER2013 test set.

TABLE VIII: Transfer Learning Results

Metric	Value
Test Accuracy	60.35%
Test Loss	1.060

The obtained performance is close to the CNN trained from scratch. However, transfer learning provides faster convergence and better feature extraction stability.

G. Fine-Tuning Strategy

After transfer learning, fine-tuning was applied in order to adapt higher-level MobileNetV2 features to the FER2013 emotion recognition task.

Instead of freezing all layers, the final layers of MobileNetV2 were unfrozen and retrained using a very small learning rate.

H. Fine-Tuning Results

After fine-tuning, the model achieved the following performance:

TABLE IX: Fine-Tuning Results

Metric	Value
Test Accuracy	61.28%
Test Loss	1.045

Fine-tuning slightly improved the classification accuracy and reduced the loss value.

I. Discussion

The experimental results demonstrate that transfer learning improves training stability and generalization.

However, the performance plateau around 61% suggests that the limitations mainly come from the dataset itself rather than the architecture alone.

The classes *happy* and *surprise* remain easier to classify due to their distinctive facial patterns, while *fear* and *disgust* remain difficult because of limited samples and visual similarity with other emotions.

Nevertheless, MobileNetV2 offers a strong compromise between:

- computational efficiency,
- inference speed,

- model size,
- classification performance.

This makes the model suitable for real-time deployment applications.

J. Comparison Between CNN and MobileNetV2

TABLE X: Comparison Between CNN and MobileNetV2

Model	Accuracy	Loss
CNN From Scratch	60.39%	1.058
MobileNetV2 Transfer Learning	60.35%	1.060
MobileNetV2 Fine-Tuning	61.28%	1.045

The results show that fine-tuned MobileNetV2 slightly outperformed the CNN developed from scratch while providing better training efficiency and faster convergence.

XII. RESNET50 TRANSFER LEARNING

After experimenting with MobileNetV2, a second transfer learning approach was implemented using the ResNet50 architecture.

ResNet50 is a deep convolutional neural network introduced by Microsoft Research. It is based on residual learning, which allows very deep neural networks to train efficiently without suffering from vanishing gradient problems.

Compared to lightweight architectures such as MobileNetV2, ResNet50 provides:

- deeper feature extraction,
- stronger representation learning,
- better hierarchical facial feature modeling,
- improved robustness for complex image classification tasks.

The objective of using ResNet50 was to evaluate whether a deeper architecture could improve facial emotion recognition performance on the FER2013 dataset.

A. Residual Learning

The main innovation of ResNet50 is the use of residual blocks.

Instead of learning a direct mapping:

$$H(x) \quad (9)$$

the network learns a residual function:

$$F(x) = H(x) - x \quad (10)$$

which gives:

$$H(x) = F(x) + x \quad (11)$$

This residual connection helps gradients propagate through very deep networks and improves training stability.

B. Transfer Learning Strategy

The ResNet50 transfer learning pipeline consisted of the following steps:

- 1) Loading the pre-trained ResNet50 model
- 2) Using ImageNet weights
- 3) Removing the original classification layer
- 4) Freezing the convolutional base
- 5) Adding custom dense layers
- 6) Training the classification head

The model was initialized using:

$$weights = ImageNet \quad (12)$$

with:

$$include_top = False \quad (13)$$

This configuration allows the network to reuse learned ImageNet visual representations while adapting the final layers to facial emotion recognition.

C. Input Preprocessing

ResNet50 requires RGB images as input.

Therefore, FER2013 grayscale images were:

- converted from grayscale to RGB,
- resized to 224×224 ,
- normalized using ResNet50 preprocessing.

The larger image size allows deeper convolutional layers to capture more detailed facial patterns.

D. Data Augmentation

To improve model generalization and reduce overfitting, several augmentation techniques were applied:

- Random horizontal flipping
- Random rotation
- Random zoom
- Random translation
- Random contrast adjustment

These transformations increase dataset diversity and improve robustness to facial variations and lighting conditions.

E. ResNet50 Architecture

The proposed ResNet50 transfer learning architecture is summarized in Table XI.

TABLE XI: ResNet50 Transfer Learning Architecture

Layer	Description
Input	$224 \times 224 \times 3$ RGB image
Data Augmentation	Random transformations
Preprocessing	ResNet50 preprocessing
ResNet50	Pre-trained convolutional base
GlobalAveragePooling2D	Feature reduction
BatchNormalization	Stabilization
Dropout	Regularization
Dense	256 neurons, ReLU
Dropout	Regularization
Dense	7 neurons, Softmax

F. Training Configuration

The ResNet50 model was trained using:

- Adam optimizer,
- sparse categorical crossentropy,
- batch size = 32,
- early stopping,
- ReduceLROnPlateau callbacks.

The initial learning rate was set to:

$$lr = 10^{-4} \quad (14)$$

A smaller learning rate was selected because deeper networks are more sensitive to large parameter updates.

G. Fine-Tuning

After the initial transfer learning phase, fine-tuning was applied.

The final layers of ResNet50 were unfrozen and retrained using a very small learning rate:

$$lr = 10^{-5} \quad (15)$$

Fine-tuning allows the higher-level ImageNet features to adapt more specifically to facial emotion recognition.

H. ResNet50 Results

The ResNet50 transfer learning model achieved the following performance on the FER2013 test dataset.

TABLE XII: ResNet50 Evaluation Results

Metric	Value
Test Accuracy	63.84%
Test Loss	0.982

The ResNet50 model outperformed both the CNN trained from scratch and the MobileNetV2 architecture.

The deeper residual architecture allowed better extraction of complex facial emotion features.

I. Discussion

The experimental results demonstrate that deeper pre-trained architectures such as ResNet50 improve facial emotion recognition performance.

Compared to MobileNetV2, ResNet50 provides:

- richer feature extraction,
- better emotion discrimination,
- improved classification stability.

However, ResNet50 also introduces:

- higher computational cost,
- larger model size,
- slower inference speed.

Despite these limitations, ResNet50 achieved the best overall performance among all tested models.

J. Comparison Between All Models

TABLE XIII: Comparison Between All Models

Model	Accuracy	Loss
CNN From Scratch	60.39%	1.058
MobileNetV2 Transfer Learning	60.35%	1.060
MobileNetV2 Fine-Tuning	61.28%	1.045
ResNet50 Transfer Learning	63.84%	0.982

The comparison shows that ResNet50 achieved the highest accuracy and the lowest loss value among all evaluated architectures.

XIII. DEPLOYMENT

The final system was deployed using Streamlit and OpenCV. The application performs the following operations in real time:

- 1) Capture webcam frames
- 2) Detect faces
- 3) Recognize known individuals
- 4) Predict emotions
- 5) Display results on screen

XIV. CONCLUSION

This project demonstrates the effectiveness of Deep Learning and Transfer Learning in face recognition and facial emotion analysis.

The proposed system successfully performs real-time face recognition and emotion prediction using convolutional neural networks and pre-trained models.

Future improvements may include:

- Larger datasets
- Better emotion classification
- GPU optimization
- Multi-face recognition

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